

59. How many ways are there to pack nine identical DVDs into three indistinguishable boxes so that each box contains at least two DVDs?
60. How many ways are there to distribute five balls into seven boxes if each box must have at most one ball in it if
- both the balls and boxes are labeled?
 - the balls are labeled, but the boxes are unlabeled?
 - the balls are unlabeled, but the boxes are labeled?
 - both the balls and boxes are unlabeled?
61. How many ways are there to distribute five balls into three boxes if each box must have at least one ball in it if
- both the balls and boxes are labeled?
 - the balls are labeled, but the boxes are unlabeled?
 - the balls are unlabeled, but the boxes are labeled?
 - both the balls and boxes are unlabeled?
62. Suppose that a basketball league has 32 teams, split into two conferences of 16 teams each. Each conference is split into three divisions. Suppose that the North Central Division has five teams. Each of the teams in the North Central Division plays four games against each of the other teams in this division, three games against each of the 11 remaining teams in the conference, and two games against each of the 16 teams in the other conference. In how many different orders can the games of one of the teams in the North Central Division be scheduled?
- *63. Suppose that a weapons inspector must inspect each of five different sites twice, visiting one site per day. The inspector is free to select the order in which to visit these sites, but cannot visit site X, the most suspicious site, on two consecutive days. In how many different orders can the inspector visit these sites?
64. How many different terms are there in the expansion of $(x_1 + x_2 + \cdots + x_m)^n$ after all terms with identical sets of exponents are added?
- *65. Prove the **Multinomial Theorem**: If n is a positive integer, then
- $$(x_1 + x_2 + \cdots + x_m)^n = \sum_{n_1 + n_2 + \cdots + n_m = n} C(n; n_1, n_2, \dots, n_m) x_1^{n_1} x_2^{n_2} \cdots x_m^{n_m},$$
- where
- $$C(n; n_1, n_2, \dots, n_m) = \frac{n!}{n_1! n_2! \cdots n_m!}$$
- is a **multinomial coefficient**.
66. Find the expansion of $(x + y + z)^4$.
67. Find the coefficient of $x^3 y^2 z^5$ in $(x + y + z)^{10}$.
68. How many terms are there in the expansion of $(x + y + z)^{100}$?

6.6

Generating Permutations and Combinations

6.6.1 Introduction

Methods for counting various types of permutations and combinations were described in the previous sections of this chapter, but sometimes permutations or combinations need to be generated, not just counted. Consider the following three problems. First, suppose that a salesperson must visit six different cities. In which order should these cities be visited to minimize total travel time? One way to determine the best order is to determine the travel time for each of the $6! = 720$ different orders in which the cities can be visited and choose the one with the smallest travel time. Second, suppose we are given a set of six positive integers and wish to find a subset of them that has 100 as their sum, if such a subset exists. One way to find these numbers is to generate all $2^6 = 64$ subsets and check the sum of their elements. Third, suppose a laboratory has 95 employees. A group of 12 of these employees with a particular set of 25 skills is needed for a project. (Each employee can have one or more of these skills.) One way to find such a set of employees is to generate all sets of 12 of these employees and check whether they have the desired skills. These examples show that it is often necessary to generate permutations and combinations to solve problems.

6.6.2 Generating Permutations

Any set with n elements can be placed in one-to-one correspondence with the set $\{1, 2, 3, \dots, n\}$. We can list the permutations of any set of n elements by generating the permutations of the n smallest positive integers and then replacing these integers with the corresponding elements. Many different algorithms have been developed to generate the $n!$ permutations of this set. We

will describe one of these that is based on the **lexicographic** (or **dictionary**) **ordering** of the set of permutations of $\{1, 2, 3, \dots, n\}$. In this ordering, the permutation $a_1 a_2 \dots a_n$ precedes the permutation of $b_1 b_2 \dots b_n$, if for some k , with $1 \leq k \leq n$, $a_1 = b_1, a_2 = b_2, \dots, a_{k-1} = b_{k-1}$, and $a_k < b_k$. In other words, a permutation of the set of the n smallest positive integers precedes (in lexicographic order) a second permutation if the number in this permutation in the first position where the two permutations disagree is smaller than the number in that position in the second permutation.

EXAMPLE 1 The permutation 23415 of the set $\{1, 2, 3, 4, 5\}$ precedes the permutation 23514, because these permutations agree in the first two positions, but the number in the third position in the first permutation, 4, is smaller than the number in the third position in the second permutation, 5. Similarly, the permutation 41532 precedes 52143. 

An algorithm for generating the permutations of $\{1, 2, \dots, n\}$ can be based on a procedure that constructs the next permutation in lexicographic order following a given permutation $a_1 a_2 \dots a_n$. We will show how this can be done. First, suppose that $a_{n-1} < a_n$. Interchange a_{n-1} and a_n to obtain a larger permutation. No other permutation is both larger than the original permutation and smaller than the permutation obtained by interchanging a_{n-1} and a_n . For instance, the next larger permutation after 234156 is 234165. On the other hand, if $a_{n-1} > a_n$, then a larger permutation cannot be obtained by interchanging these last two terms in the permutation. Look at the last three integers in the permutation. If $a_{n-2} < a_{n-1}$, then the last three integers in the permutation can be rearranged to obtain the next largest permutation. Put the smaller of the two integers a_{n-1} and a_n that is greater than a_{n-2} in position $n - 2$. Then, place the remaining integer and a_{n-2} into the last two positions in increasing order. For instance, the next larger permutation after 234165 is 234516.



On the other hand, if $a_{n-2} > a_{n-1}$ (and $a_{n-1} > a_n$), then a larger permutation cannot be obtained by permuting the last three terms in the permutation. Based on these observations, a general method can be described for producing the next larger permutation in increasing order following a given permutation $a_1 a_2 \dots a_n$. First, find the integers q and q_{+1} with $q < q_{+1}$ and

$$q_{+1} > q_{+2} > \dots > a_n,$$

that is, the last pair of adjacent integers in the permutation where the first integer in the pair is smaller than the second. Then, the next larger permutation in lexicographic order is obtained by putting in the j th position the least integer among $q_{+1}, q_{+2}, \dots$, and a_n that is greater than q and listing in increasing order the rest of the integers $q, q_{+1}, \dots, a_n$ in positions $j + 1$ to n . It is easy to see that there is no other permutation larger than the permutation $a_1 a_2 \dots a_n$ but smaller than the new permutation produced. (The verification of this fact is left as an exercise for the reader.)

EXAMPLE 2 What is the next permutation in lexicographic order after 362541?

Extra Examples 

Solution: The last pair of integers q and q_{+1} where $q < q_{+1}$ is $a_3 = 2$ and $a_4 = 5$. The least integer to the right of 2 that is greater than 2 in the permutation is $a_5 = 4$. Hence, 4 is placed in the third position. Then the integers 2, 5, and 1 are placed in order in the last three positions, giving 125 as the last three positions of the permutation. Hence, the next permutation is 364125. 

To produce the $n!$ permutations of the integers $1, 2, 3, \dots, n$, begin with the smallest permutation in lexicographic order, namely, $123 \dots n$, and successively apply the procedure described for producing the next larger permutation of $n! - 1$ times. This yields all the permutations of the n smallest integers in lexicographic order.

EXAMPLE 3 Generate the permutations of the integers 1, 2, 3 in lexicographic order.

Solution: Begin with 123. The next permutation is obtained by interchanging 3 and 2 to obtain 132. Next, because $3 > 2$ and $1 < 3$, permute the three integers in 132. Put the smaller of 3 and 2 in the first position, and then put 1 and 3 in increasing order in positions 2 and 3 to obtain 213. This is followed by 231, obtained by interchanging 1 and 3, because $1 < 3$. The next larger permutation has 3 in the first position, followed by 1 and 2 in increasing order, namely, 312. Finally, interchange 1 and 2 to obtain the last permutation, 321. We have generated the permutations of 1, 2, 3 in lexicographic order. They are 123, 132, 213, 231, 312, and 321. ◀

Algorithm 1 displays the procedure for finding the next permutation in lexicographic order after a permutation that is not $n \ n - 1 \ n - 2 \ \dots \ 2 \ 1$, which is the largest permutation.

ALGORITHM 1 Generating the Next Permutation in Lexicographic Order.

```

procedure next permutation( $a_1 a_2 \dots a_n$ ; permutation of
    {1, 2, ...,  $n$ } not equal to  $n \ n - 1 \ \dots \ 2 \ 1$ )
     $j := n - 1$ 
    while  $a_j > a_{j+1}$ 
         $j := j - 1$ 
    { $j$  is the largest subscript with  $a_j < a_{j+1}$ }
     $k := n$ 
    while  $a_j > a_k$ 
         $k := k - 1$ 
    { $a_k$  is the smallest integer greater than  $a_j$  to the right of  $a_j$ }
    interchange  $a_j$  and  $a_k$ 
     $r := n$ 
     $s := j + 1$ 
    while  $r > s$ 
        interchange  $a_r$  and  $a_s$ 
         $r := r - 1$ 
         $s := s + 1$ 
    {this puts the tail end of the permutation after the  $j$ th position in increasing order}
    { $a_1 a_2 \dots a_n$  is now the next permutation}

```

6.6.3 Generating Combinations



How can we generate all the combinations of the elements of a finite set? Because a combination is just a subset, we can use the correspondence between subsets of $\{a_1, a_2, \dots, a_n\}$ and bit strings of length n .

Recall that the bit string corresponding to a subset has a 1 in position k if a_k is in the subset, and has a 0 in this position if a_k is not in the subset. If all the bit strings of length n can be listed, then by the correspondence between subsets and bit strings, a list of all the subsets is obtained.

Recall that a bit string of length n is also the binary expansion of an integer between 0 and $2^n - 1$. The 2^n bit strings can be listed in order of their increasing size as integers in their binary expansions. To produce all binary expansions of length n , start with the bit string 000 ... 00, with n zeros. Then, successively find the next expansion until the bit string 111 ... 11 is obtained. At each stage the next binary expansion is found by locating the first position from the right that is not a 1, then changing all the 1s to the right of this position to 0s and making this first 0 (from the right) a 1.

EXAMPLE 4 Find the next bit string after 10 0010 0111.

Solution: The first bit from the right that is not a 1 is the fourth bit from the right. Change this bit to a 1 and change all the following bits to 0s. This produces the next larger bit string, 10 0010 1000. ◀

The procedure for producing the next larger bit string after $b_{n-1}b_{n-2}\dots b_1b_0$ is given as Algorithm 2.

ALGORITHM 2 Generating the Next Larger Bit String.

```

procedure next bit string( $b_{n-1} b_{n-2} \dots b_1 b_0$ : bit string not equal to 11...11)
 $i := 0$ 
while  $b_i = 1$ 
     $b_i := 0$ 
     $i := i + 1$ 
 $b_i := 1$ 
 $\{b_{n-1} b_{n-2} \dots b_1 b_0$  is now the next bit string}
  
```

Next, an algorithm for generating the r -combinations of the set $\{1, 2, 3, \dots, n\}$ will be given. An r -combination can be represented by a sequence containing the elements in the subset in increasing order. The r -combinations can be listed using lexicographic order on these sequences. In this lexicographic ordering, the first r -combination is $\{1, 2, \dots, r-1, r\}$ and the last r -combination is $\{n-r+1, n-r+2, \dots, n-1, n\}$. The next r -combination after $a_1 a_2 \dots a_r$ can be obtained in the following way: First, locate the last element a_i in the sequence such that $a_i \neq n-r+i$. Then, replace a_i with $a_i + 1$ and a_j with $a_i + j - i + 1$, for $j = i+1, i+2, \dots, r$. It is left for the reader to show that this produces the next larger r -combination in lexicographic order. This procedure is illustrated with Example 5.

EXAMPLE 5 Find the next larger 4-combination of the set $\{1, 2, 3, 4, 5, 6\}$ after $\{1, 2, 5, 6\}$.

Solution: The last term among the terms a_i with $a_1 = 1$, $a_2 = 2$, $a_3 = 5$, and $a_4 = 6$ such that $a_i \neq 6 - 4 + i$ is $a_2 = 2$. To obtain the next larger 4-combination, increment a_2 by 1 to obtain $a_2 = 3$. Then set $a_3 = 3 + 1 = 4$ and $a_4 = 3 + 2 = 5$. Hence the next larger 4-combination is $\{1, 3, 4, 5\}$. ◀

Algorithm 3 displays pseudocode for this procedure.

ALGORITHM 3 Generating the Next r -Combination in Lexicographic Order.

```

procedure next  $r$ -combination( $\{a_1, a_2, \dots, a_r\}$ : proper subset of
     $\{1, 2, \dots, n\}$  not equal to  $\{n-r+1, \dots, n\}$  with
     $a_1 < a_2 < \dots < a_r$ )
 $i := r$ 
while  $a_i = n - r + i$ 
     $i := i - 1$ 
 $a_i := a_i + 1$ 
for  $j := i + 1$  to  $r$ 
     $a_j := a_i + j - i$ 
 $\{\{a_1, a_2, \dots, a_r\}$  is now the next combination}
  
```

Exercises

- Place these permutations of $\{1, 2, 3, 4, 5\}$ in lexicographic order: 43521, 15432, 45321, 23451, 23514, 14532, 21345, 45213, 31452, 31542.
- Place these permutations of $\{1, 2, 3, 4, 5, 6\}$ in lexicographic order: 234561, 231456, 165432, 156423, 543216, 541236, 231465, 314562, 432561, 654321, 654312, 435612.
- The name of a file in a computer directory consists of three uppercase letters followed by a digit, where each letter is either A, B, or C, and each digit is either 1 or 2. List the name of these files in lexicographic order, where we order letters using the usual alphabetic order of letters.
- Suppose that the name of a file in a computer directory consists of three digits followed by two lowercase letters and each digit is 0, 1, or 2, and each letter is either a or b. List the name of these files in lexicographic order, where we order letters using the usual alphabetic order of letters.
- Find the next larger permutation in lexicographic order after each of these permutations.

a) 1432	b) 54123	c) 12453
d) 45231	e) 6714235	f) 31528764
- Find the next larger permutation in lexicographic order after each of these permutations.

a) 1342	b) 45321	c) 13245
d) 612345	e) 1623547	f) 23587416
- Use Algorithm 1 to generate the 24 permutations of the first four positive integers in lexicographic order.
- Use Algorithm 2 to list all the subsets of the set $\{1, 2, 3, 4\}$.
- Use Algorithm 3 to list all the 3-combinations of $\{1, 2, 3, 4, 5\}$.
- Show that Algorithm 1 produces the next larger permutation in lexicographic order.
- Show that Algorithm 3 produces the next larger r -combination in lexicographic order after a given r -combination.
- Develop an algorithm for generating the r -permutations of a set of n elements.
- List all 3-permutations of $\{1, 2, 3, 4, 5\}$.

The remaining exercises in this section develop another algorithm for generating the permutations of $\{1, 2, 3, \dots, n\}$. This algorithm is based on Cantor expansions of integers. Every nonnegative integer less than $n!$ has a unique Cantor expansion

$$a_1 1! + a_2 2! + \dots + a_{n-1} (n-1)!$$

where a_i is a nonnegative integer not exceeding i , for $i = 1, 2, \dots, n-1$. The integers $a_1, a_2, \dots, a_{n-1}$ are called the **Cantor digits** of this integer.

Given a permutation of $\{1, 2, \dots, n\}$, let $a_{k-1}, k = 2, 3, \dots, n$, be the number of integers less than k that follow k in the permutation. For instance, in the permutation 43215, a_1 is the number of integers less than 2 that follow 2, so $a_1 = 1$. Similarly, for this example $a_2 = 2$, $a_3 = 3$, and $a_4 = 0$. Consider the function from the set of permutations of $\{1, 2, 3, \dots, n\}$ to the set of nonnegative integers less than $n!$ that sends a permutation to the integer that has $a_1, a_2, \dots, a_{n-1}$, defined in this way, as its Cantor digits.

- Find the Cantor digits $a_1, a_2, \dots, a_{n-1}$ that correspond to these permutations.

a) 246531	b) 12345	c) 654321
-----------	----------	-----------
- Show that the correspondence described in the preamble is a bijection between the set of permutations of $\{1, 2, 3, \dots, n\}$ and the nonnegative integers less than $n!$.
- Find the permutations of $\{1, 2, 3, 4, 5\}$ that correspond to these integers with respect to the correspondence between Cantor expansions and permutations as described in the preamble to Exercise 14.

a) 3	b) 89	c) 111
------	-------	--------
- Develop an algorithm for producing all permutations of a set of n elements based on the correspondence described in the preamble to Exercise 14.

Key Terms and Results

TERMS

combinatorics: the study of arrangements of objects

enumeration: the counting of arrangements of objects

tree diagram: a diagram made up of a root, branches leaving the root, and other branches leaving some of the endpoints of branches

permutation: an ordered arrangement of the elements of a set

r -permutation: an ordered arrangement of r elements of a set

$P(n, r)$: the number of r -permutations of a set with n elements

r -combination: an unordered selection of r elements of a set

$C(n, r)$: the number of r -combinations of a set with n elements

binomial coefficient $\binom{n}{r}$: also the number of r -combinations of a set with n elements

combinatorial proof: a proof that uses counting arguments rather than algebraic manipulation to prove a result

Pascal's triangle: a representation of the binomial coefficients where the i th row of the triangle contains $\binom{i}{j}$ for $j = 0, 1, 2, \dots, i$

$S(n, j)$: the Stirling number of the second kind denoting the number of ways to distribute n distinguishable objects into j indistinguishable boxes so that no box is empty

RESULTS

product rule for counting: The number of ways to do a procedure that consists of two tasks is the product of the number of ways to do the first task and the number of ways to do the second task after the first task has been done.

product rule for sets: The number of elements in the Cartesian product of finite sets is the product of the number of elements in each set.

sum rule for counting: The number of ways to do a task in one of two ways is the sum of the number of ways to do these tasks if they cannot be done simultaneously.

sum rule for sets: The number of elements in the union of pairwise disjoint finite sets is the sum of the numbers of elements in these sets.

subtraction rule for counting or inclusion–exclusion for sets: If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

subtraction rule or inclusion–exclusion for sets: The number of elements in the union of two sets is the sum of the number of elements in these sets minus the number of elements in their intersection.

division rule for counting: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

division rule for sets: Suppose that a finite set A is the union of n disjoint subsets each with d elements. Then $n = |A|/d$.

the pigeonhole principle: When more than k objects are placed in k boxes, there must be a box containing more than one object.

the generalized pigeonhole principle: When N objects are placed in k boxes, there must be a box containing at least $\lceil N/k \rceil$ objects.

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Pascal's identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

the binomial theorem: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$

There are n^r r -permutations of a set with n elements when repetition is allowed.

There are $C(n+r-1, r)$ r -combinations of a set with n elements when repetition is allowed.

There are $n!/(n_1!n_2!\cdots n_k!)$ permutations of n objects of k types where there are n_i indistinguishable objects of type i for $i = 1, 2, 3, \dots, k$.

the algorithm for generating the permutations of the set $\{1, 2, \dots, n\}$

Review Questions

1. Explain how the sum and product rules can be used to find the number of bit strings with a length not exceeding 10.
2. Explain how to find the number of bit strings of length not exceeding 10 that have at least one 0 bit.
3. a) How can the product rule be used to find the number of functions from a set with m elements to a set with n elements?
b) How many functions are there from a set with five elements to a set with 10 elements?
c) How can the product rule be used to find the number of one-to-one functions from a set with m elements to a set with n elements?
d) How many one-to-one functions are there from a set with five elements to a set with 10 elements?
e) How many onto functions are there from a set with five elements to a set with 10 elements?
4. How can you find the number of possible outcomes of a playoff between two teams where the first team that wins four games wins the playoff?
5. How can you find the number of bit strings of length ten that either begin with 101 or end with 010?
6. a) State the pigeonhole principle.
b) Explain how the pigeonhole principle can be used to show that among any 11 integers, at least two must have the same last digit.
7. a) State the generalized pigeonhole principle.
b) Explain how the generalized pigeonhole principle can be used to show that among any 91 integers, there are at least ten that end with the same digit.
8. a) What is the difference between an r -combination and an r -permutation of a set with n elements?
b) Derive an equation that relates the number of r -combinations and the number of r -permutations of a set with n elements.
c) How many ways are there to select six students from a class of 25 to serve on a committee?
d) How many ways are there to select six students from a class of 25 to hold six different executive positions on a committee?
9. a) What is Pascal's triangle?
b) How can a row of Pascal's triangle be produced from the one above it?
10. What is meant by a combinatorial proof of an identity? How is such a proof different from an algebraic one?
11. Explain how to prove Pascal's identity using a combinatorial argument.

12. a) State the binomial theorem.
 b) Explain how to prove the binomial theorem using a combinatorial argument.
 c) Find the coefficient of $x^{100}y^{101}$ in the expansion of $(2x + 5y)^{201}$.
13. a) Explain how to find a formula for the number of ways to select r objects from n objects when repetition is allowed and order does not matter.
 b) How many ways are there to select a dozen objects from among objects of five different types if objects of the same type are indistinguishable?
 c) How many ways are there to select a dozen objects from these five different types if there must be at least three objects of the first type?
 d) How many ways are there to select a dozen objects from these five different types if there cannot be more than four objects of the first type?
 e) How many ways are there to select a dozen objects from these five different types if there must be at least two objects of the first type, but no more than three objects of the second type?
14. a) Let n and r be positive integers. Explain why the number of solutions of the equation $x_1 + x_2 + \cdots + x_n = r$, where x_i is a nonnegative integer for $i = 1, 2, 3, \dots, n$, equals the number of r -combinations of a set with n elements.
 b) How many solutions in nonnegative integers are there to the equation $x_1 + x_2 + x_3 + x_4 = 17$?
 c) How many solutions in positive integers are there to the equation in part (b)?
15. a) Derive a formula for the number of permutations of n objects of k different types, where there are n_1 indistinguishable objects of type one, n_2 indistinguishable objects of type two, $\dots$, and n_k indistinguishable objects of type k .
 b) How many ways are there to order the letters of the word *INDISCREETNESS*?
16. Describe an algorithm for generating all the permutations of the set of the n smallest positive integers.
17. a) How many ways are there to deal hands of five cards to six players from a standard 52-card deck?
 b) How many ways are there to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed in box i ?
18. Describe an algorithm for generating all the combinations of the set of the n smallest positive integers.

Supplementary Exercises

1. How many ways are there to choose 6 items from 10 distinct items when
 - a) the items in the choices are ordered and repetition is not allowed?
 - b) the items in the choices are ordered and repetition is allowed?
 - c) the items in the choices are unordered and repetition is not allowed?
 - d) the items in the choices are unordered and repetition is allowed?
2. How many ways are there to choose 10 items from 6 distinct items when
 - a) the items in the choices are ordered and repetition is not allowed?
 - b) the items in the choices are ordered and repetition is allowed?
 - c) the items in the choices are unordered and repetition is not allowed?
 - d) the items in the choices are unordered and repetition is allowed?
3. A test contains 100 true/false questions. How many different ways can a student answer the questions on the test, if answers may be left blank?
4. How many strings of length 10 either start with 000 or end with 1111?
5. How many bit strings of length 10 over the alphabet $\{a, b, c\}$ have either exactly three a s or exactly four b s?
6. The internal telephone numbers in the phone system on a campus consist of five digits, with the first digit not equal to zero. How many different numbers can be assigned in this system?
7. An ice cream parlor has 28 different flavors, 8 different kinds of sauce, and 12 toppings.
 - a) In how many different ways can a dish of three scoops of ice cream be made where each flavor can be used more than once and the order of the scoops does not matter?
 - b) How many different kinds of small sundaes are there if a small sundae contains one scoop of ice cream, a sauce, and a topping?
 - c) How many different kinds of large sundaes are there if a large sundae contains three scoops of ice cream, where each flavor can be used more than once and the order of the scoops does not matter; two kinds of sauce, where each sauce can be used only once and the order of the sauces does not matter; and three toppings, where each topping can be used only once and the order of the toppings does not matter?
8. How many positive integers less than 1000
 - a) have exactly three decimal digits?
 - b) have an odd number of decimal digits?
 - c) have at least one decimal digit equal to 9?
 - d) have no odd decimal digits?
 - e) have two consecutive decimal digits equal to 5?
 - f) are palindromes (that is, read the same forward and backward)?

9. When the numbers from 1 to 1000 are written out in decimal notation, how many of each of these digits are used?
 a) 0 b) 1 c) 2 d) 9
10. There are 12 signs of the zodiac. How many people are needed to guarantee that at least six of these people have the same sign?
11. A fortune cookie company makes 213 different fortunes. A student eats at a restaurant that uses fortunes from this company and gives each customer one fortune cookie at the end of a meal. What is the largest possible number of times that the student can eat at the restaurant without getting the same fortune four times?
12. How many people are needed to guarantee that at least two were born on the same day of the week and in the same month (perhaps in different years)?
13. Show that given any set of 10 positive integers not exceeding 50 there exist at least two different five-element subsets of this set that have the same sum.
14. A package of baseball cards contains 20 cards. How many packages must be purchased to ensure that two cards in these packages are identical if there are a total of 550 different cards?
15. a) How many cards must be chosen from a standard deck of 52 cards to guarantee that at least two of the four aces are chosen?
 b) How many cards must be chosen from a standard deck of 52 cards to guarantee that at least two of the four aces and at least two of the 13 kinds are chosen?
 c) How many cards must be chosen from a standard deck of 52 cards to guarantee that there are at least two cards of the same kind?
 d) How many cards must be chosen from a standard deck of 52 cards to guarantee that there are at least two cards of each of two different kinds?
- *16. Show that in any set of $n + 1$ positive integers not exceeding $2n$ there must be two that are relatively prime.
- *17. Show that in a sequence of m integers there exists one or more consecutive terms with a sum divisible by m .
18. Show that if five points are picked in the interior of a square with a side length of 2, then at least two of these points are no farther than $\sqrt{2}$ apart.
19. Show that the decimal expansion of a rational number must repeat itself from some point onward.
20. Once a computer worm infects a personal computer via an infected e-mail message, it sends a copy of itself to 100 e-mail addresses it finds in the electronic message mailbox on this personal computer. What is the maximum number of different computers this one computer can infect in the time it takes for the infected message to be forwarded five times?
21. How many ways are there to choose a dozen donuts from 20 varieties
 a) if there are no two donuts of the same variety?
 b) if all donuts are of the same variety?
 c) if there are no restrictions?
- d) if there are at least two varieties among the dozen donuts chosen?
 e) if there must be at least six blueberry-filled donuts?
 f) if there can be no more than six blueberry-filled donuts?
22. Find n if
 a) $P(n, 2) = 110$. b) $P(n, n) = 5040$.
 c) $P(n, 4) = 12P(n, 2)$.
23. Find n if
 a) $C(n, 2) = 45$. b) $C(n, 3) = P(n, 2)$.
 c) $C(n, 5) = C(n, 2)$.
24. Show that if n and r are nonnegative integers and $n \geq r$, then

$$P(n + 1, r) = P(n, r)(n + 1)/(n + 1 - r).$$
- *25. Suppose that S is a set with n elements. How many ordered pairs (A, B) are there such that A and B are subsets of S with $A \subseteq B$? [Hint: Show that each element of S belongs to A , $B - A$, or $S - B$.]
26. Give a combinatorial proof of Corollary 2 of Section 6.4 by setting up a correspondence between the subsets of a set with an even number of elements and the subsets of this set with an odd number of elements. [Hint: Take an element a in the set. Set up the correspondence by putting a in the subset if it is not already in it and taking it out if it is in the subset.]
27. Let n and r be integers with $1 \leq r < n$. Show that

$$C(n, r - 1) = C(n + 2, r + 1) - 2C(n + 1, r + 1) + C(n, r + 1).$$
28. Prove using mathematical induction that $\sum_{j=2}^n C(j, 2) = C(n + 1, 3)$ whenever n is an integer greater than 1.
29. Show that if n is an integer then

$$\sum_{k=0}^n 3^k \binom{n}{k} = 4^n.$$
30. Show that $\sum_{i=1}^{n-1} \sum_{j=i+1}^n 1 = \binom{n}{2}$ if n is an integer with $n \geq 2$.
31. Show that $\sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n 1 = \binom{n}{3}$ if n is an integer with $n \geq 3$.
32. In this exercise we will derive a formula for the sum of the squares of the n smallest positive integers. We will count the number of triples (i, j, k) where i, j , and k are integers such that $0 \leq i < k$, $0 \leq j < k$, and $1 \leq k \leq n$ in two ways.
 a) Show that there are k^2 such triples with a fixed k . Deduce that there are $\sum_{k=1}^n k^2$ such triples.
 b) Show that the number of such triples with $0 \leq i < j < k$ and the number of such triples with $0 \leq j < i < k$ both equal $C(n + 1, 3)$.
 c) Show that the number of such triples with $0 \leq i = j < k$ equals $C(n + 1, 2)$.

- d) Combining part (a) with parts (b) and (c), conclude that
- $$\sum_{k=1}^n k^2 = 2C(n+1, 3) + C(n+1, 2)$$
- $$= n(n+1)(2n+1)/6.$$
- *33. How many bit strings of length n , where $n \geq 4$, contain exactly two occurrences of 01?
34. Let S be a set. We say that a collection of subsets $A_1, A_2, \dots, A_n$ each containing d elements, where $d \geq 2$, is *2-colorable* if it is possible to assign to each element of S one of two different colors so that in every subset A_i there are elements that have been assigned each color. Let $m(d)$ be the largest integer such that every collection of fewer than $m(d)$ sets each containing d elements is 2-colorable.
- a) Show that the collection of all subsets with d elements of a set S with $2d - 1$ elements is not 2-colorable.
- b) Show that $m(2) = 3$.
- **c) Show that $m(3) = 7$. [Hint: Show that the collection $\{1, 3, 5\}$, $\{1, 2, 6\}$, $\{1, 4, 7\}$, $\{2, 3, 4\}$, $\{2, 5, 7\}$, $\{3, 6, 7\}$, $\{4, 5, 6\}$ is not 2-colorable. Then show that all collections of six sets with three elements each are 2-colorable.]
35. A professor writes 20 multiple-choice questions, each with the possible answer a, b, c , or d , for a discrete mathematics test. If the number of questions with a, b, c , and d as their answer is 8, 3, 4, and 5, respectively, how many different answer keys are possible, if the questions can be placed in any order?
36. How many different arrangements are there of eight people seated at a round table, where two arrangements are considered the same if one can be obtained from the other by a rotation?
37. How many ways are there to assign 24 students to five faculty advisors?
38. How many ways are there to choose a dozen apples from a bushel containing 20 indistinguishable Delicious apples, 20 indistinguishable Macintosh apples, and 20 indistinguishable Granny Smith apples, if at least three of each kind must be chosen?
39. How many solutions are there to the equation $x_1 + x_2 + x_3 = 17$, where x_1, x_2 , and x_3 are nonnegative integers with
- a) $x_1 > 1, x_2 > 2$, and $x_3 > 3$?
b) $x_1 < 6$ and $x_3 > 5$?
c) $x_1 < 4, x_2 < 3$, and $x_3 > 5$?
40. a) How many different strings can be made from the word PEPPERCORN when all the letters are used?
b) How many of these strings start and end with the letter P?
c) In how many of these strings are the three letter Ps consecutive?
41. How many subsets of a set with ten elements
- a) have fewer than five elements?
b) have more than seven elements?
c) have an odd number of elements?
42. A witness to a hit-and-run accident tells the police that the license plate of the car in the accident, which contains three letters followed by three digits, starts with the letters AS and contains both the digits 1 and 2. How many different license plates can fit this description?
43. How many ways are there to put n identical objects into m distinct containers so that no container is empty?
44. How many ways are there to seat six boys and eight girls in a row of chairs so that no two boys are seated next to each other?
45. How many ways are there to distribute six objects to five boxes if
- a) both the objects and boxes are labeled?
b) the objects are labeled, but the boxes are unlabeled?
c) the objects are unlabeled, but the boxes are labeled?
d) both the objects and the boxes are unlabeled?
46. How many ways are there to distribute five objects into six boxes if
- a) both the objects and boxes are labeled?
b) the objects are labeled, but the boxes are unlabeled?
c) the objects are unlabeled, but the boxes are labeled?
d) both the objects and the boxes are unlabeled?
- The **signless Stirling number of the first kind** $c(n, k)$, where k and n are integers with $1 \leq k \leq n$, equals the number of ways to arrange n people around k circular tables with at least one person seated at each table, where two seatings of m people around a circular table are considered the same if everyone has the same left neighbor and the same right neighbor.
47. Find these signless Stirling numbers of the first kind.
- a) $c(3, 2)$ b) $c(4, 2)$
c) $c(4, 3)$ d) $c(5, 4)$
48. Show that if n is a positive integer, then $\sum_{j=1}^n c(n, j) = n!$.
49. Show that if n is a positive integer with $n \geq 3$, then $c(n, n-2) = (3n-1)C(n, 3)/4$.
- *50. Show that if n and k are integers with $1 \leq k < n$, then $c(n+1, k) = c(n, k-1) + nc(n, k)$.
51. Give a combinatorial proof that 2^n divides $n!$ whenever n is an even positive integer. [Hint: Use Theorem 3 in Section 6.5 to count the number of permutations of $2n$ objects where there are two indistinguishable objects of n different types.]
52. How many 11-element RNA sequences consist of 4 As, 3 Cs, 2 Us, and 2 Gs, and end with CAA?
- Exercises 53 and 54 are based on a discussion in [RoTe09]. A method used in the 1960s for sequencing RNA chains used enzymes to break chains after certain links. Some enzymes break RNA chains after each G link, while others break them after each C or U link. Using these enzymes it is sometimes possible to correctly sequence all the bases in an RNA chain.
- *53. Suppose that when an enzyme that breaks RNA chains after each G link is applied to a 12-link chain, the fragments obtained are G, CCG, AAAG, and UCCG, and when an enzyme that breaks RNA chains after each C or U link is applied, the fragments obtained are C, C, C, C, GGU,

The **signless Stirling number of the first kind** $c(n, k)$, where k and n are integers with $1 \leq k \leq n$, equals the number of ways to arrange n people around k circular tables with at least one person seated at each table, where two seatings of m people around a circular table are considered the same if everyone has the same left neighbor and the same right neighbor.

47. Find these signless Stirling numbers of the first kind.

a) $c(3, 2)$	b) $c(4, 2)$
c) $c(4, 3)$	d) $c(5, 4)$
 48. Show that if n is a positive integer, then $\sum_{j=1}^n c(n, j) = n!$.
 49. Show that if n is a positive integer with $n \geq 3$, then $c(n, n-2) = (3n-1)C(n, 3)/4$.
 50. Show that if n and k are integers with $1 \leq k < n$, then $c(n+1, k) = c(n, k-1) + nc(n, k)$.
 51. Give a combinatorial proof that 2^n divides $n!$ whenever n is an even positive integer. [Hint: Use Theorem 3 in Section 6.5 to count the number of permutations of $2n$ objects where there are two indistinguishable objects of n different types.]
 52. How many 11-element RNA sequences consist of 4 As, 3 Cs, 2 Us, and 2 Gs, and end with CAA?
- Exercises 53 and 54 are based on a discussion in [RoTe09]. A method used in the 1960s for sequencing RNA chains used enzymes to break chains after certain links. Some enzymes break RNA chains after each G link, while others break them after each C or U link. Using these enzymes it is sometimes possible to correctly sequence all the bases in an RNA chain.
53. Suppose that when an enzyme that breaks RNA chains after each G link is applied to a 12-link chain, the fragments obtained are G, CCG, AAAG, and UCCG, and when an enzyme that breaks RNA chains after each C or U link is applied, the fragments obtained are C, C, C, C, GGU,

and GAAAG. Can you determine the entire 12-link RNA chain from these two sets of fragments? If so, what is this RNA chain?

- *54. Suppose that when an enzyme that breaks RNA chains after each G link is applied to a 12-link chain, the fragments obtained are AC, UG, and ACG and when an enzyme that breaks RNA chains after each C or U link is applied, the fragments obtained are U, GAC, and GAC. Can you determine the entire RNA chain from these two sets of fragments? If so, what is this RNA chain?
55. Devise an algorithm for generating all the r -permutations of a finite set when repetition is allowed.
56. Devise an algorithm for generating all the r -combinations of a finite set when repetition is allowed.
- *57. Show that if m and n are integers with $m \geq 3$ and $n \geq 3$, then $R(m, n) \leq R(m, n-1) + R(m-1, n)$.
- *58. Show that $R(3, 4) \geq 7$ by showing that in a group of six people, where any two people are friends or enemies, there are not necessarily three mutual friends or four mutual enemies.

Computer Projects

Write programs with these input and output.

- Given a positive integer n and a nonnegative integer not exceeding n , find the number of r -permutations and r -combinations of a set with n elements.
- Given positive integers n and r , find the number of r -permutations when repetition is allowed and r -combinations when repetition is allowed of a set with n elements.
- Given a sequence of positive integers, find the longest increasing and the longest decreasing subsequence of the sequence.
- *4. Given an equation $x_1 + x_2 + \cdots + x_n = C$, where C is a constant, and $x_1, x_2, \dots, x_n$ are nonnegative integers, list all the solutions.
- Given a positive integer n , list all the permutations of the set $\{1, 2, 3, \dots, n\}$ in lexicographic order.
- Given a positive integer n and a nonnegative integer r not exceeding n , list all the r -combinations of the set $\{1, 2, 3, \dots, n\}$ in lexicographic order.
- Given a positive integer n and a nonnegative integer r not exceeding n , list all the r -permutations of the set $\{1, 2, 3, \dots, n\}$ in lexicographic order.
- Given a positive integer n , list all the combinations of the set $\{1, 2, 3, \dots, n\}$.
- Given positive integers n and r , list all the r -permutations, with repetition allowed, of the set $\{1, 2, 3, \dots, n\}$.
- Given positive integers n and r , list all the r -combinations, with repetition allowed, of the set $\{1, 2, 3, \dots, n\}$.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

- Find the number of possible outcomes in a two-team play-off when the winner is the first team to win 5 out of 9, 6 out of 11, 7 out of 13, and 8 out of 15.
- Which binomial coefficients are odd? Can you formulate a conjecture based on numerical evidence?
- Verify that $C(2n, n)$ is divisible by the square of a prime, when $n \neq 1, 2$, or 4, for as many positive integers n as you can. [That $C(2n, n)$ is divisible by the square of a prime with $n \neq 1, 2$, or 4 was proved in 1996 by Andrew Granville and Olivier Ramaré, settling a conjecture made in 1980 by Paul Erdős and Ron Graham.]
- Find as many odd integers n less than 200 as you can for which $C(n, \lfloor n/2 \rfloor)$ is not divisible by the square of a prime. Formulate a conjecture based on your evidence.
- *5. For each integer less than 100 determine whether $C(2n, n)$ is divisible by 3. Can you formulate a conjecture that tells us for which integers n the binomial coefficient $C(2n, n)$ is divisible by 3 based on the digits in the base three expansion of n ?
- Generate all the permutations of a set with eight elements.
- Generate all the 6-permutations of a set with nine elements.
- Generate all combinations of a set with eight elements.
- Generate all 5-combinations with repetition allowed of a set with seven elements.

Writing Projects

Respond to these with essays using outside sources.

1. Describe some of the earliest uses of the pigeonhole principle by Dirichlet and other mathematicians.
2. Discuss ways in which the current telephone numbering plan can be extended to accommodate the rapid demand for more telephone numbers. (See if you can find some of the proposals coming from the telecommunications industry.) For each new numbering plan you discuss, show how to find the number of different telephone numbers it supports.
3. Discuss the importance of combinatorial reasoning in gene sequencing and related problems involving genomes.
4. Many combinatorial identities are described in this book. Find some sources of such identities and describe important combinatorial identities besides those already introduced in this book. Give some representative proofs, including combinatorial ones, of some of these identities.
5. Describe the different models used to model the distribution of particles in statistical mechanics, including Maxwell–Boltzmann, Bose–Einstein, and Fermi–Dirac statistics. In each case, describe the counting techniques used in the model.
6. Define the Stirling numbers of the first kind and describe some of their properties and the identities they satisfy.
7. Describe some of the properties and the identities that Stirling numbers of the second kind satisfy, including the connection between Stirling numbers of the first and second kinds.
8. Describe the latest discoveries of values and bounds for Ramsey numbers.
9. Describe additional ways to generate all the permutations of a set with n elements besides those found in Section 6.6. Compare these algorithms and the algorithms described in the text and exercises of Section 6.6 in terms of their computational complexity.
10. Describe at least one way to generate all the partitions of a positive integer n . (See Exercise 49 in Section 5.3.)

